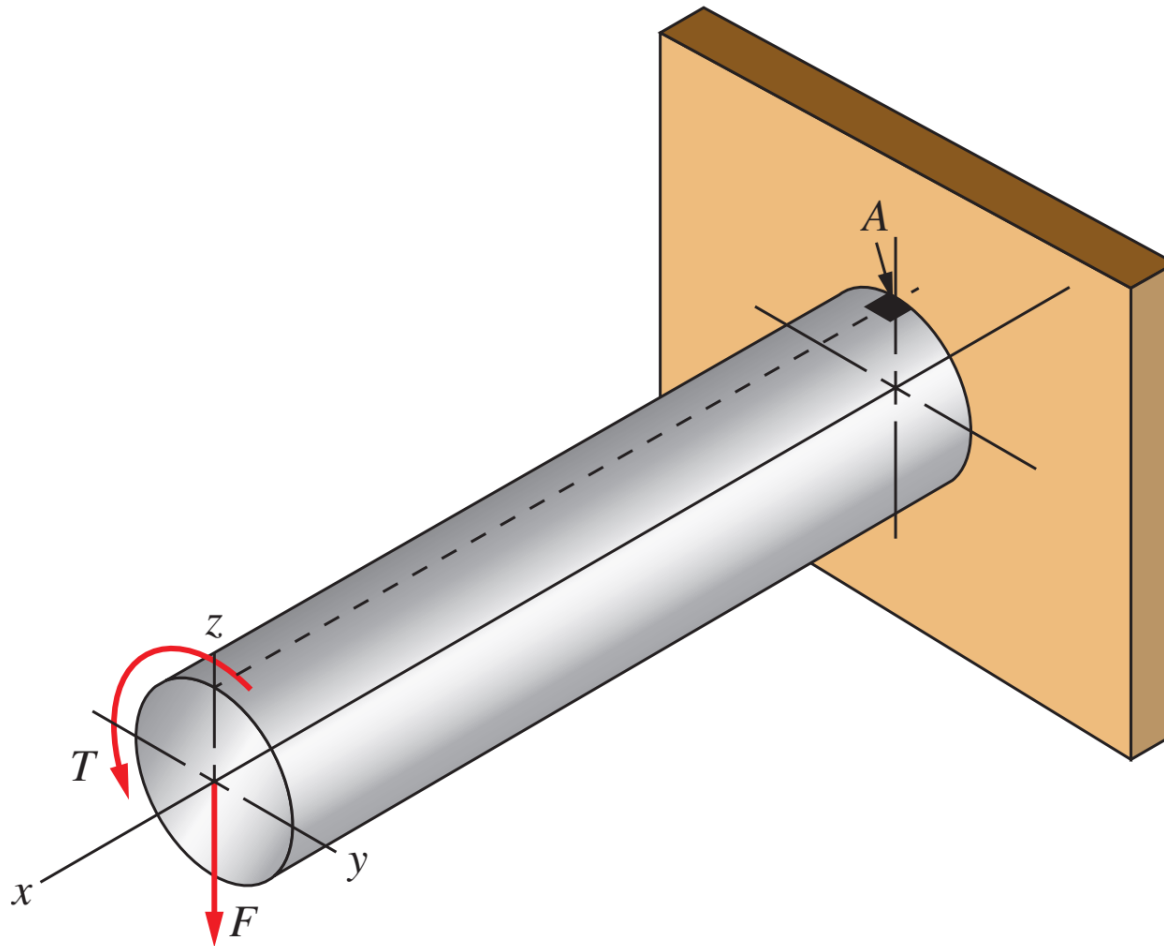


1. A circular bar with diameter 2 inch and length of 20 inch is fixed at one end and then is loaded at the other end by a downward force of 2 kip and a torsional moment of 30 kip $\cdot$ in. Determine the absolute maximum shear stress at the top of the shaft (point A). (**10'** in total)



Solutions to Problem #1:

The downward force causes bending in the bar with the maximum moment at the point A where the bar is attached to the support. The bending moment causes a tensile stress on the top of the bar in the x -direction at the point A , where the magnitude of the stress is

$$\sigma_x = M/S \quad (S \text{ is the section modulus of the round bar})$$

The torsional moment causes torsional shear stress in the xy plane at point A having a magnitude of

$$\tau_{xy} = T/Z_p \quad (Z_p \text{ is the polar section modulus of the round bar})$$

Point A then is subjected to a tensile stress combined with shear. This special case can be shown in the Mohr's circle.

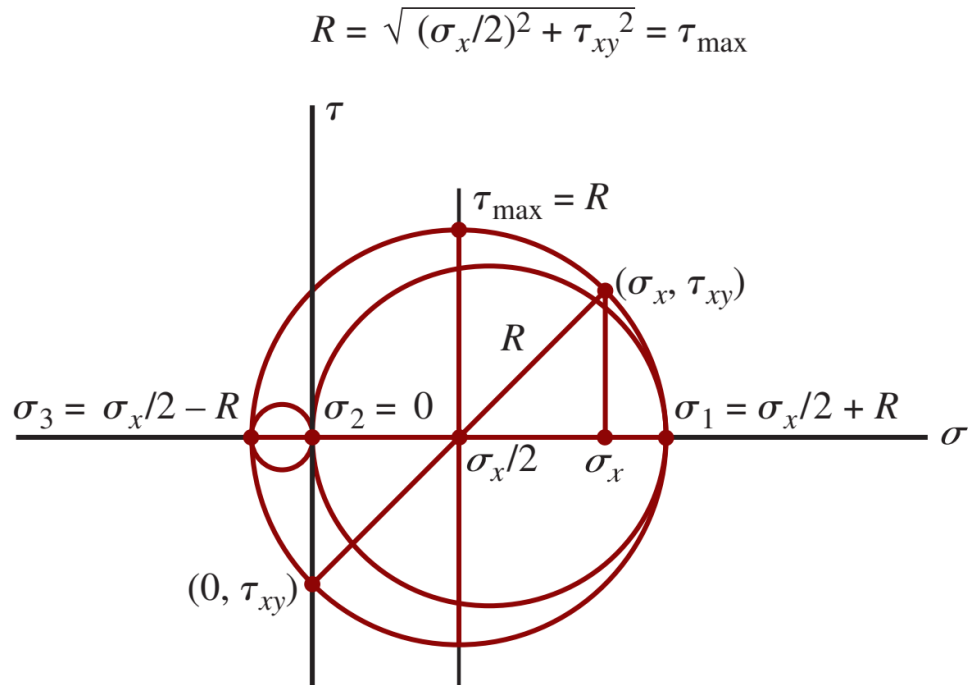
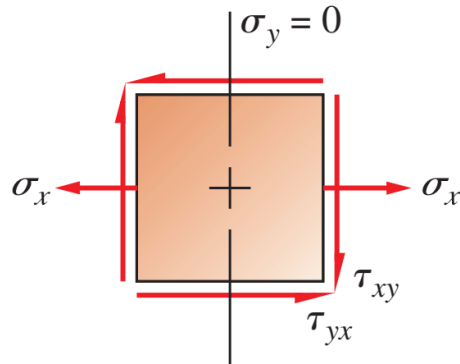
Solutions to Problem #1, cont'd:

The absolute maximum shear stress is then

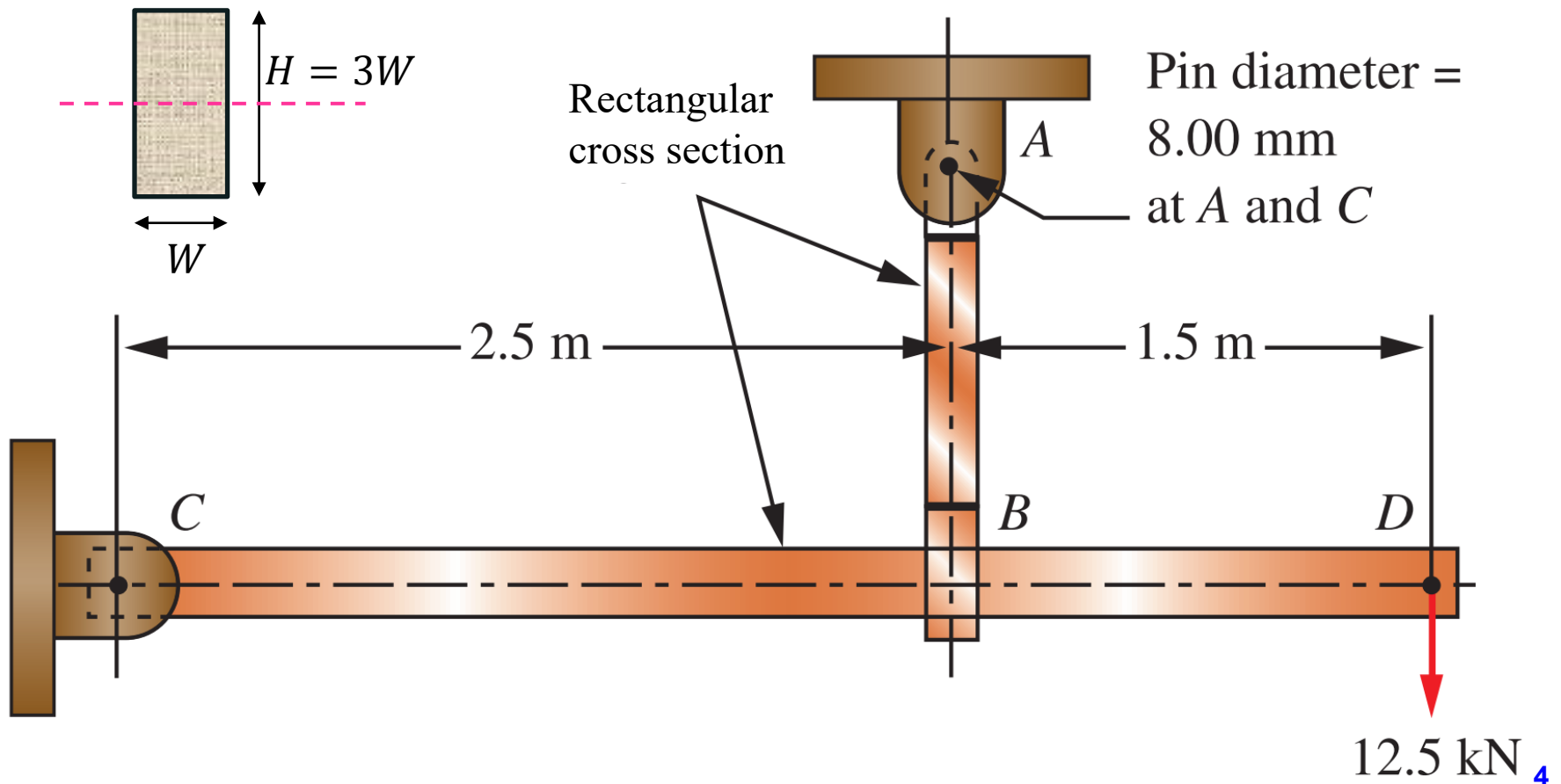
$$\tau_{\max}^{\text{abs}} = R = \sqrt{(\sigma_x/2)^2 + (\tau_{xy})^2} = \sqrt{(M/2S)^2 + (T/Z_p)^2}$$

For round bar, $Z_p = 2S$, then

$$\tau_{\max}^{\text{abs}} = \frac{\sqrt{M^2 + T^2}}{Z_p} = \frac{\sqrt{(2 \times 20000)^2 + (30000)^2}}{\pi 2^3 / 16} = 31831 \text{ psi} = \boxed{31.8 \text{ ksi}}$$



2. A horizontal beam supported by a vertical tension link. Determine the minimum width of a rectangular cross section for member $C-D$ (the height must be 3 times of width), so that the maximum normal stress in member $C-D$ does not exceed 500 MPa. The dashed line in the cross section indicates the axis along which the bending occurs. (15' in total)



Solutions to Problem #2:

Step 1: Determine the support reactions (R_B and R_C) for the loadings by using Free-Body-Diagrams:

$$\sum F_y = 0 \quad \rightarrow \quad R_B + R_C = 12.5 \text{ kN} \quad (1)$$

$$\sum M_C = 0 \quad \rightarrow \quad R_B \times 2.5 = 12.5 \times (2.5 + 1.5) \quad (2)$$

Solving Eqs. (1) and (2) we get:

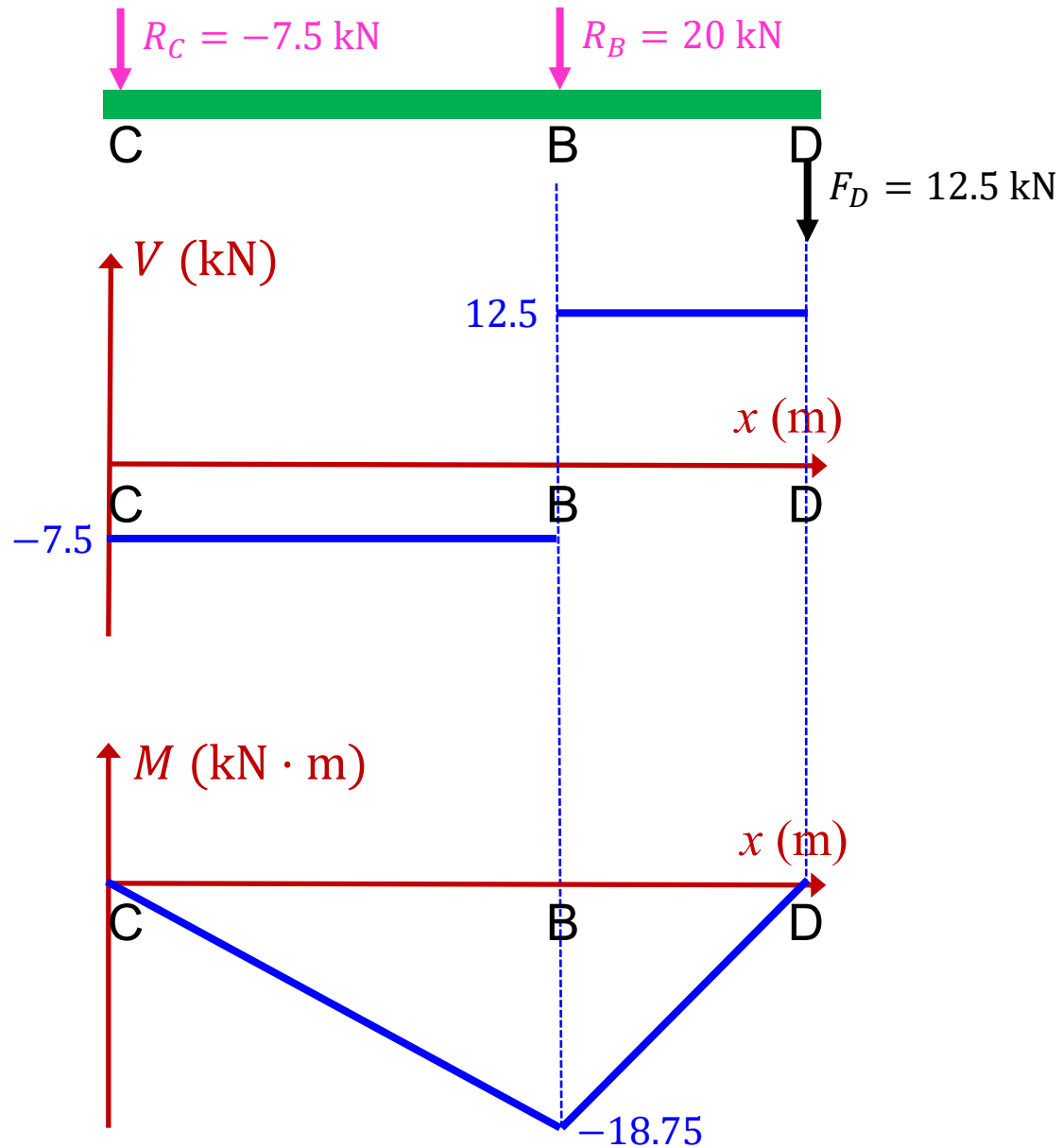
$$R_B = 20 \text{ kN}$$

$$R_C = -7.5 \text{ kN} \quad (\text{negative sign means our presumed direction should be reversed})$$

Step 2: plot the shear force and bending moment diagrams using graphical method.

From bending moment diagram, we find that the maximum bending moment happens at position B with magnitude of $18.75 \text{ kN} \cdot \text{m}$.

Solutions to Problem #2, cont'd:



Solutions to Problem #2, cont'd:

Step 3: To meet design stress $\sigma_{\max} = 500 \text{ MPa}$, the required section modulus should be at least:

$$S \geq \frac{M_{\max}}{\sigma_{\max}} = \frac{18.75 \text{ kN} \cdot \text{m}}{500 \text{ MPa}} = \frac{18.75 \times 10^3 \text{ N} \cdot \text{m}}{500 \times 10^6 \frac{\text{N}}{\text{m}^2}} = 3.75 \times 10^{-5} \text{ m}^3 = 37.5 \text{ cm}^3$$

Step 4: From Appendix 1, we know that the section modulus for a rectangular cross section

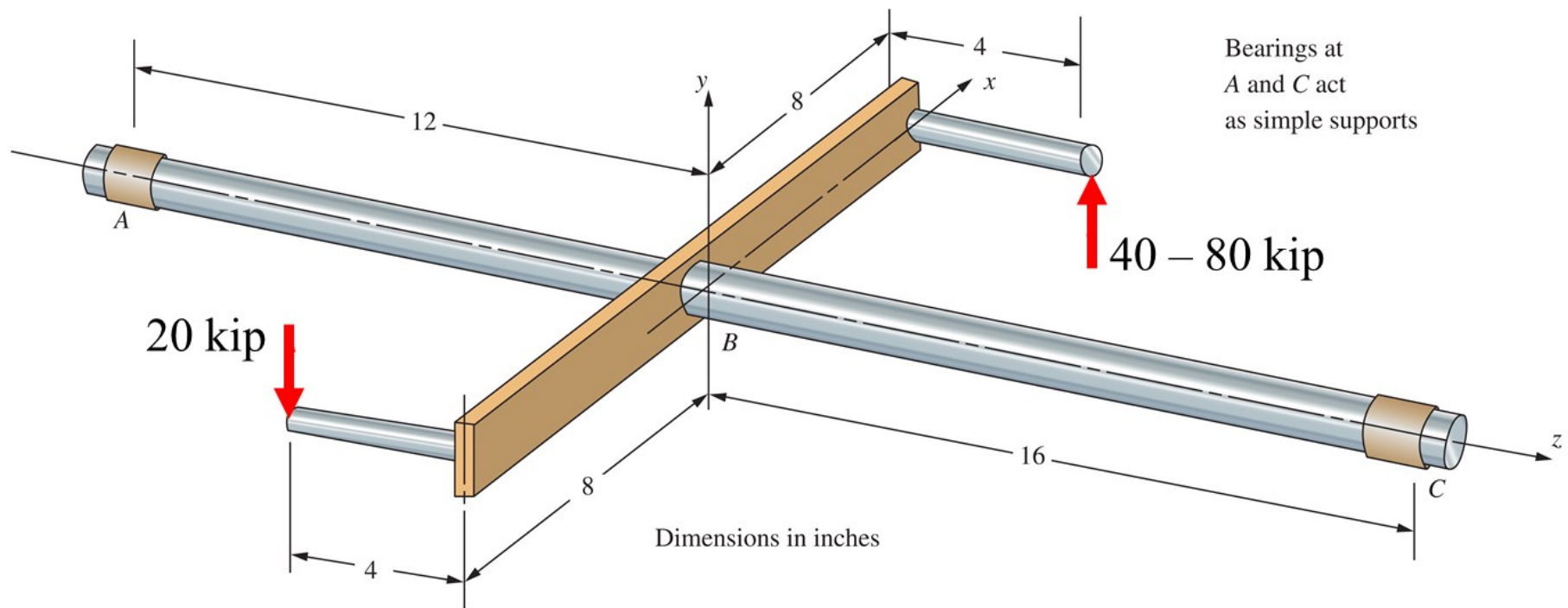
$$S = \frac{WH^2}{6} = \frac{W(3W)^2}{6} = 1.5W^3 \geq 37.5 \text{ cm}^3$$

We finally get

$$W \geq 2.92 \text{ cm}$$

$$W_{\min} = \boxed{2.92 \text{ cm}}$$

3. The diameter of the main shaft portion, labeled A , B , and C is 8.736 in. The left load 20 kip remains constantly applied, while the right load is applied and changes from 40 to 80 kip as the machine cycles. Determine the stress ratio for the point B . Also, for the same point, use the maximum shear stress theory to determine a specific material type under Aluminum alloy 2219. Take safety factor $N = 5$. (25' in total)



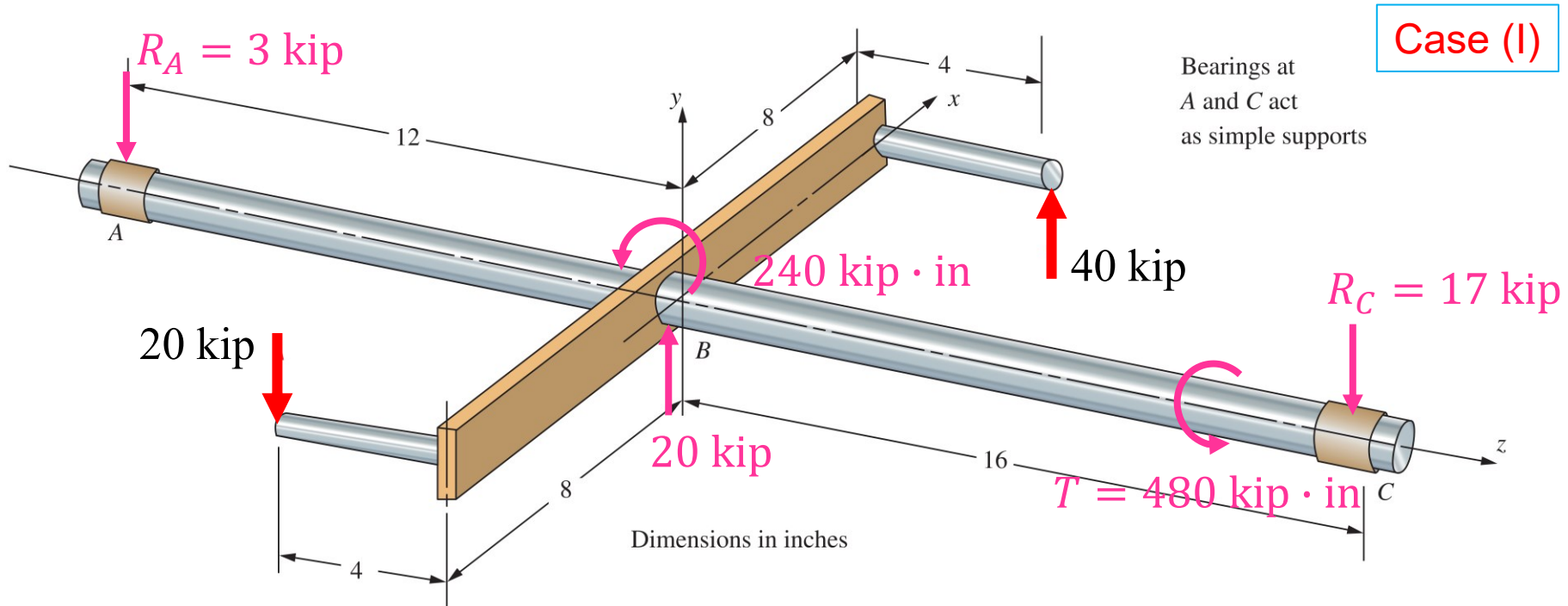
Solutions to Problem #3:

Step 1: Determine the maximum and minimum stress for the point B .

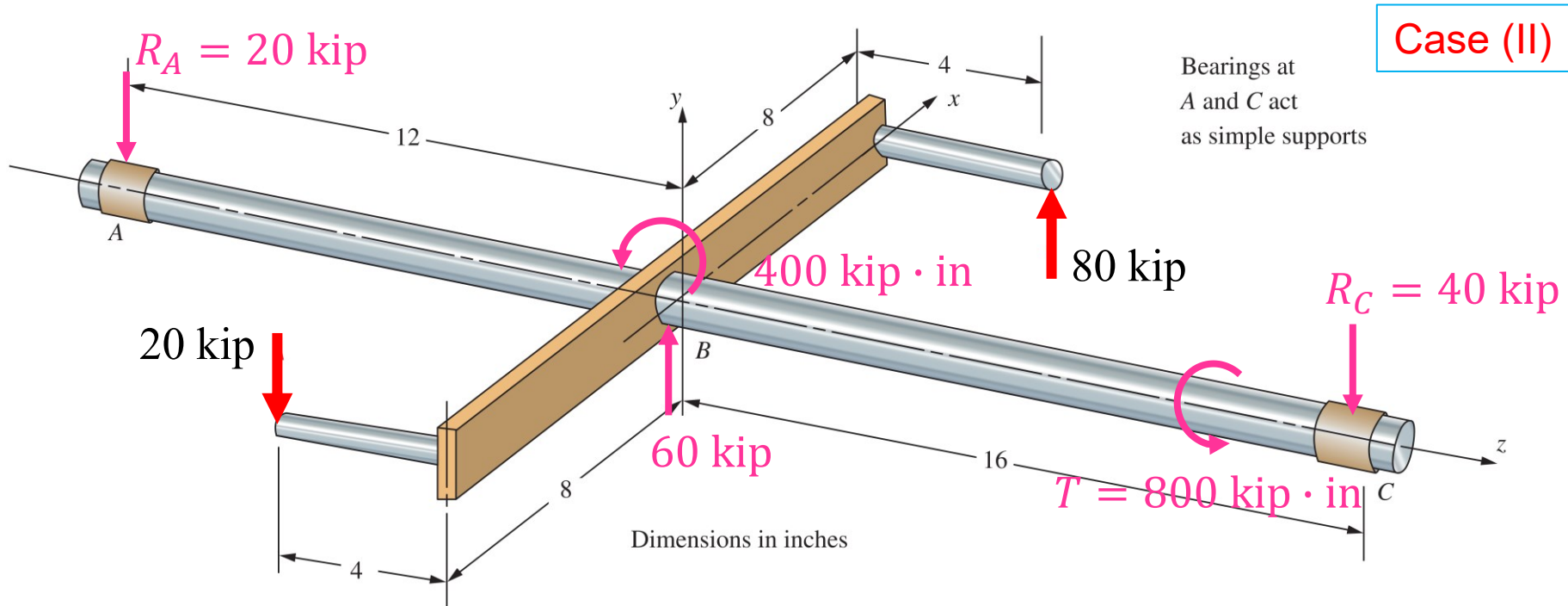
There are two loading cases: (I) $P_2 = 40$ lb; (II) $P_2 = 80$ lb, both with $P_1 = 20$ lb.

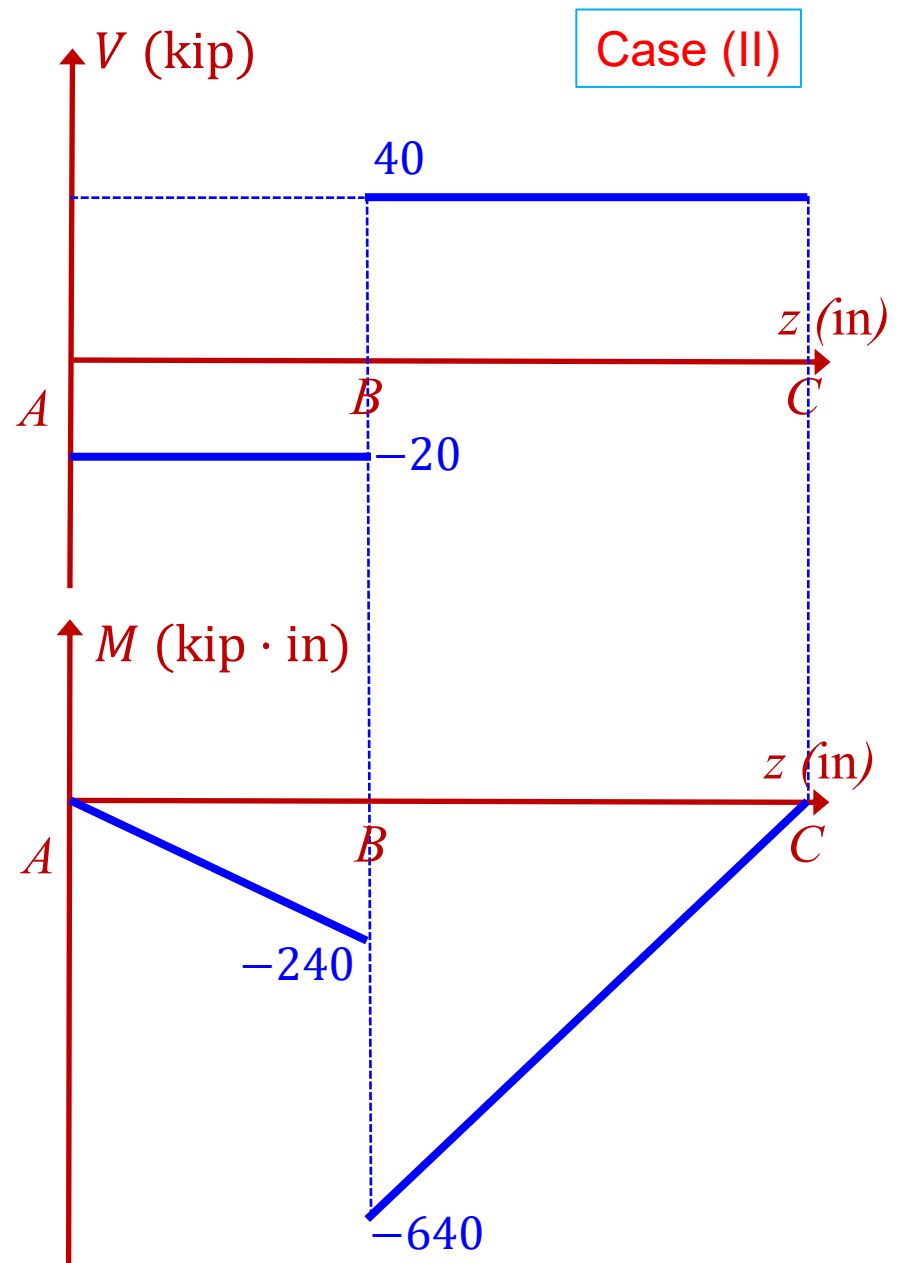
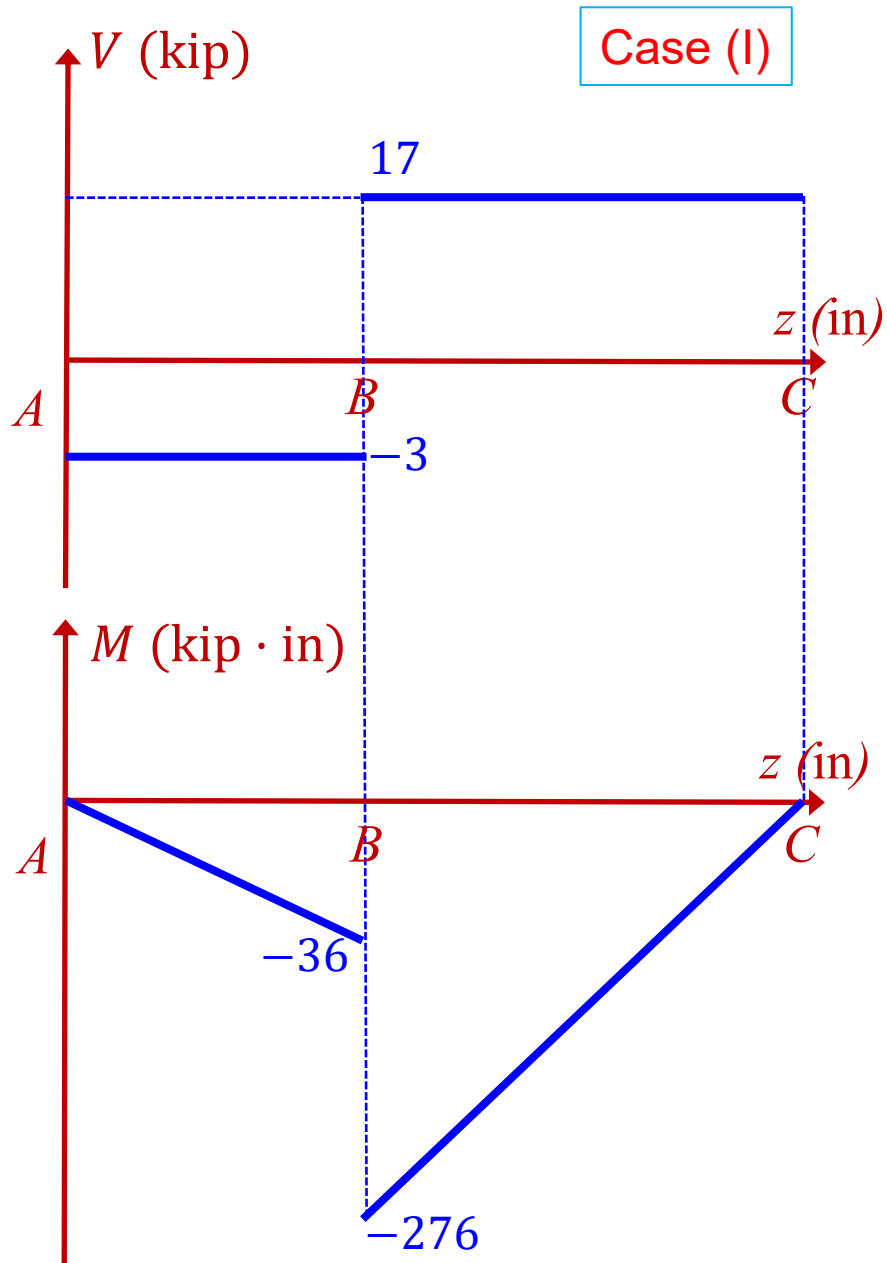
Using Free-Body-Diagrams to solve the reactions at two ends and force and moment diagrams:

Solutions to Problem #3, cont'd:



Solutions to Problem #3, cont'd:





Solutions to Problem #3, cont'd:

For the point B , the maximum bending moment for the loading case (I) and (II) are 276 kip · in and 640 kip · in, respectively.

With circular cross section, according to Appendix 1, the bending modulus

$$S = \frac{\pi D^3}{32} = \frac{\pi(8.736)^3}{32} = 65.45 \text{ in}^3$$

The maximum and minimum stresses during the cyclic loading are

$$\sigma_{\max} = \frac{M_{\max}}{S} = \frac{640000}{65.45} = 9778 \text{ psi}$$

$$\sigma_{\min} = \frac{M_{\min}}{S} = \frac{276000}{65.45} = 4217 \text{ psi}$$

$$\text{Stress ratio } R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{4217}{9778} = \boxed{0.43}$$

Solutions to Problem #3, cont'd:

Step 2: The case (II) induces the highest possible normal tensile stress on the surface of the shaft by the maximum bending moment (640 kip · in), which is also 2D plane stress state. The corresponding square stress state and Mohr's circle can be drawn.

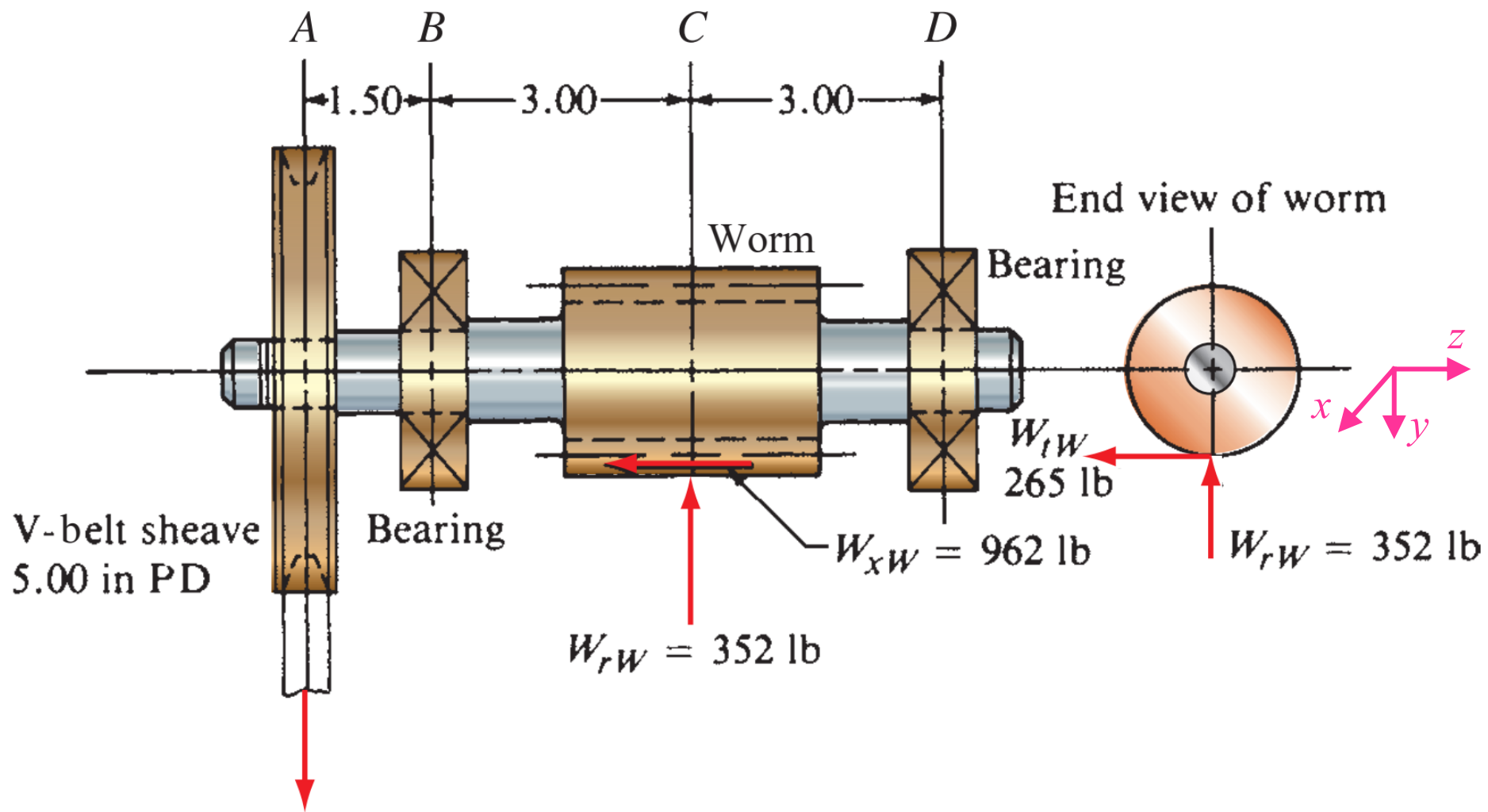
With $\sigma_x = \sigma_{\max} = 9778$ psi, $\sigma_y = \tau_{xy} = 0$, we know the two principal stresses $\sigma_1 = \sigma_x = 9778$ psi, $\sigma_2 = \sigma_3 = 0$, and the absolute maximum shear stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \frac{9778}{2} = 4889 \text{ psi}$$

The Maximum Shear Stress Theory (MSST) says that a material with tensile yield strength should fulfill $\frac{s_y}{N} \geq \sigma_1 - \sigma_3$. With $N = 5$, we get $s_y \geq N(\sigma_1 - \sigma_3) = 5(9778) = 48890 \text{ psi} = 48.89 \text{ ksi}$

From Appendix 9, we find that Aluminum alloy 2219-T87 with $s_y = 57 \text{ ksi}$ is acceptable.

4. The shaft is rotating at 1750 rpm, and it receives 7.5 hp through a 5.00-in V-belt drive from a mating belt sheave directly below. The power is delivered by a wormgear with a pitch diameter of 2.00 in. The forces on the wormgear are shown in the figure. Use SAE 1020 cold-drawn steel for the shaft material. (25' in total)
- 1) Determine the torque at A . Also determine the torque distribution in $A-B$, $B-C$, and $C-D$ region.
 - 2) Plot the shear and bending moment diagrams for the $A-B-C$ region of the shaft in the horizontal and vertical planes.
 - 3) Determine the total bending moment at point C (**for both to the left and to the right**).
 - 4) Determine the minimum acceptable diameters for the shaft at point C . **Do the design to the left and to the right of point C separately.** Take $C_s = 0.90$, $C_R = 0.81$, design factor $N = 4$, stress concentration factor $K_t = 2.0$.



Solutions to Problem #4:

Step 1: Analyze the torque distribution in the shaft.

The gear receives 7.5 hp at point A . Then, the torque at the V-belt A :

$$T_A = \frac{63000(7.5 \text{ hp})}{1750 \text{ rpm}} = \boxed{270 \text{ lb} \cdot \text{in}}$$

Since the power is delivered by a wormgear at the point C , using Free-Body-Diagram, we get torque distribution in the shaft:

$$\begin{aligned} T_{A-B} &= T_{B-C} = T_A = 270 \text{ lb} \cdot \text{in} \\ T_{C-D} &= 0 \text{ lb} \cdot \text{in} \end{aligned}$$

Solutions to Problem #4, cont'd:

Step 2: Compute forces that act in the vertical and horizontal planes.

Forces on sheave at A :

$$F_{Ay} = \frac{1.5T}{\frac{D_A}{2}} = \frac{1.5(270)}{\frac{5.00}{2}} = 162 \text{ lb} \quad (\text{downward or positive } y\text{-direction})$$

Forces on gear C :

With the given tangential, radial, and axial forces on the gear as shown in the figure, we assign them as the components of the bending forces:

$$F_{Cx} = W_{tW} = 265 \text{ lb} \quad (\text{out of plane or positive } x\text{-direction})$$

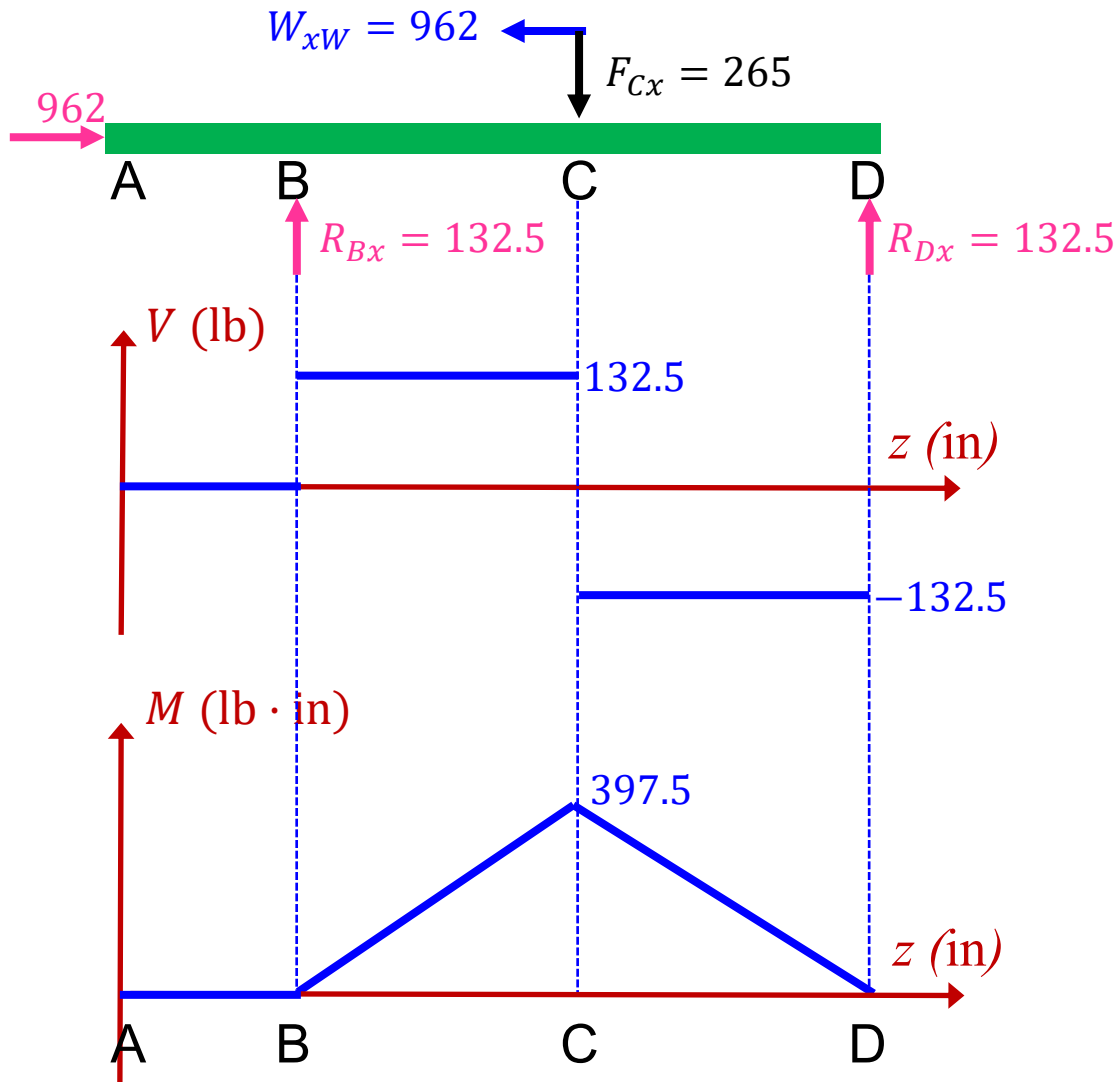
$$F_{Cy} = -W_{rB} = -352 \text{ lb} \quad (\text{upward or negative } y\text{-direction})$$

Additional bending moment at C due to the off-center axial force (W_{xW}):

$$M_C = W_{xW} \times \frac{D_C}{2} = (962 \text{ lb}) \left(\frac{2.00}{2} \text{ in} \right) = 962 \text{ lb} \cdot \text{in}$$

Solutions to Problem #4, cont'd:

Step 3: Shear force and bending moment diagrams.

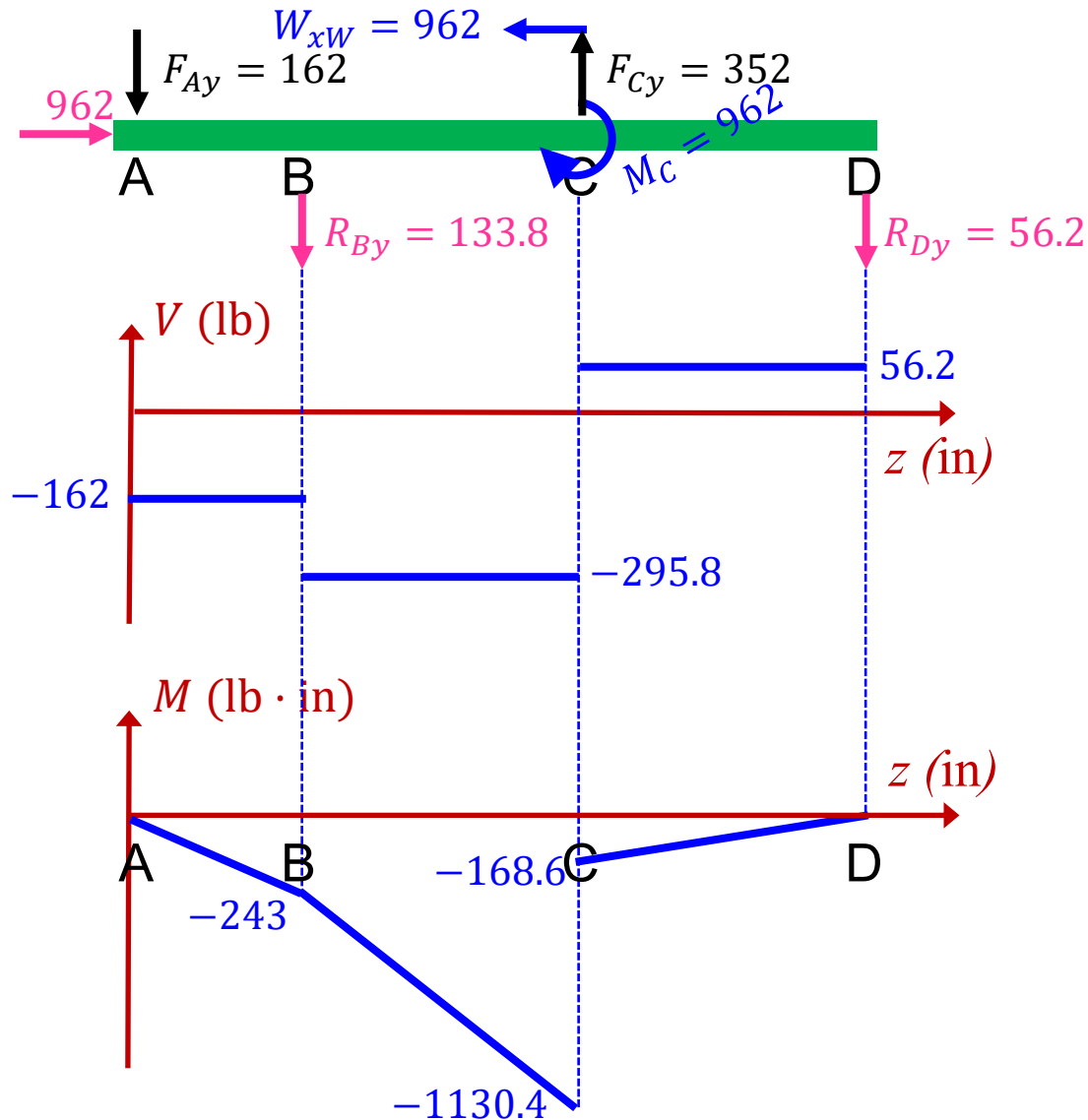


xz -plane / horizontal plane

Solutions to Problem #4, cont'd:

Step 3: Shear force and bending moment diagrams.

yz-plane / vertical plane



Solutions to Problem #4, cont'd:

Total bending moment at point C to the left:

$$M_C|_{\text{left}} = \sqrt{397.5^2 + 1130.4^2} = 1198 \text{ lb} \cdot \text{in}$$

Total bending moment at point D :

$$M_C|_{\text{right}} = \sqrt{397.5^2 + 168.6^2} = 432 \text{ lb} \cdot \text{in}$$

Solutions to Problem #4, cont'd:

Step 4: Design shaft diameter at point C.

The material is SAE 1020 cold-drawn steel. From Appendix 3, we find $s_y = 51$ ksi, $s_u = 61$ ksi

From Figure 5-11(a), we find the endurance limit $s_n = 22$ ksi. Then, the estimated actual endurance limit is

$$s'_n = s_n C_m C_{st} C_R C_S = (22)(1.0)(1.0)(0.81)(0.90) = 16.038 \text{ ksi} = 16038 \text{ psi}$$

For point C to the left (Keyseat):

$$T_C(\text{left}) = T_{A-B} = 270 \text{ lb} \cdot \text{in}, \quad M_C|_{\text{left}} = 1198 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only, with $K_t = 2.0$ and $N = 4$:

$$D_C|_{\text{left}} = \left[\frac{32(4)}{\pi} \sqrt{\left(\frac{2.0(1198)}{16038} \right)^2 + \frac{3}{4} \left(\frac{270}{51000} \right)^2} \right]^{1/3} = \boxed{1.826 \text{ in}}$$

Solutions to Problem #4, cont'd:

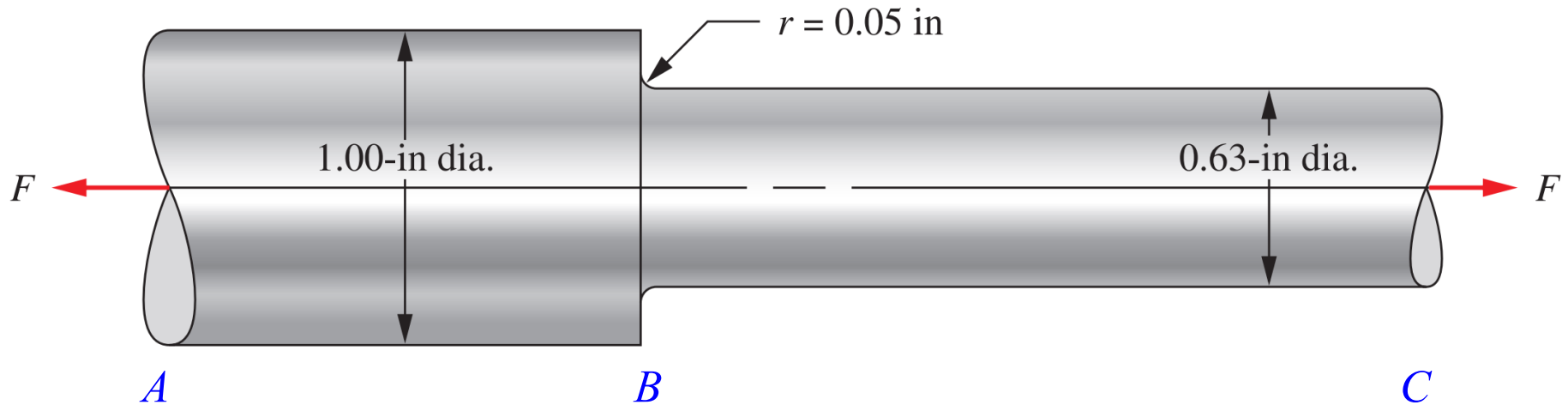
For point C to the right (Ring groove):

$$T_C(\text{right}) = T_{C-D} = 0 \text{ lb} \cdot \text{in}, \quad M_C|_{\text{right}} = 432 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only,

$$D_B|_{\text{right}} = \left[\frac{32(4)}{\pi} \sqrt{\left(\frac{2.0(432)}{16038} \right)^2 + \frac{3}{4} \left(\frac{0}{51000} \right)^2} \right]^{1/3} = \boxed{1.300 \text{ in}}$$

5. The stepped rod shown is subjected to a direct tensile force that varies from 8500 to 16 000 lb. If the rod is made from SAE 1340 OQT 700 wrought steel, use Goodman criterion to determine the design factor. Take reliability as 90%. Stress concentration needs to be considered at point *B*. (15' in total)



Solutions to Problem #5:

Step 1: Determine the maximum and minimum stress in the shaft over time.

Using Free-Body-Diagrams to solve the reactions at end A and then plot the force and moment diagrams. Note that, the load F varies from 8500 to 16000 lb. Thus, the mean and alternating loading (i.e., the amplitude of loading) is

$$F_m = \frac{8500 + 16000}{2} = 12250 \text{ lb}$$

$$F_a = \frac{16000 - 8500}{2} = 3750 \text{ lb}$$

Then, the corresponding mean and alternating (tensile) stress is

$$\sigma_m = \frac{F_m}{S} = \frac{12250}{\pi \times 0.63^3 / 4} = 39298 \text{ psi}$$

$$\sigma_a = \frac{F_a}{S} = \frac{3750}{\pi \times 0.63^3 / 4} = 12030 \text{ psi}$$

Solutions to Problem #5, cont'd:

Step 2: Material SAE 1340 OQT 700: $s_y = 197$ ksi, $s_u = 221$ ksi

From Figure 5-11, we find $s_n = 65$ ksi

Use Eq. (5-21) in the textbook to determine s_n' :

$$s_n' = s_n \cdot C_m \cdot C_{st} \cdot C_R \cdot C_s = 65(1.0)(0.80)(0.90)(1.0) = 46.8 \text{ ksi}$$

Wrought steel: $C_m = 1.0$

Axial loading: $C_{st} = 0.80$

Reliability factor: $C_R = 0.90$ (reliability 90%, see Table 5-3)

Size factor: $C_s = 1.0$ (any shape in repeated direct axial tensile stress)

Solutions to Problem #5, cont'd:

Step 3: At fillets (point B) determine stress concentration factor:

$$r/d = 0.05/0.63 = 0.079 \approx 0.08$$

$$D/d = 1.00/0.63 = 1.59 \approx 1.6$$

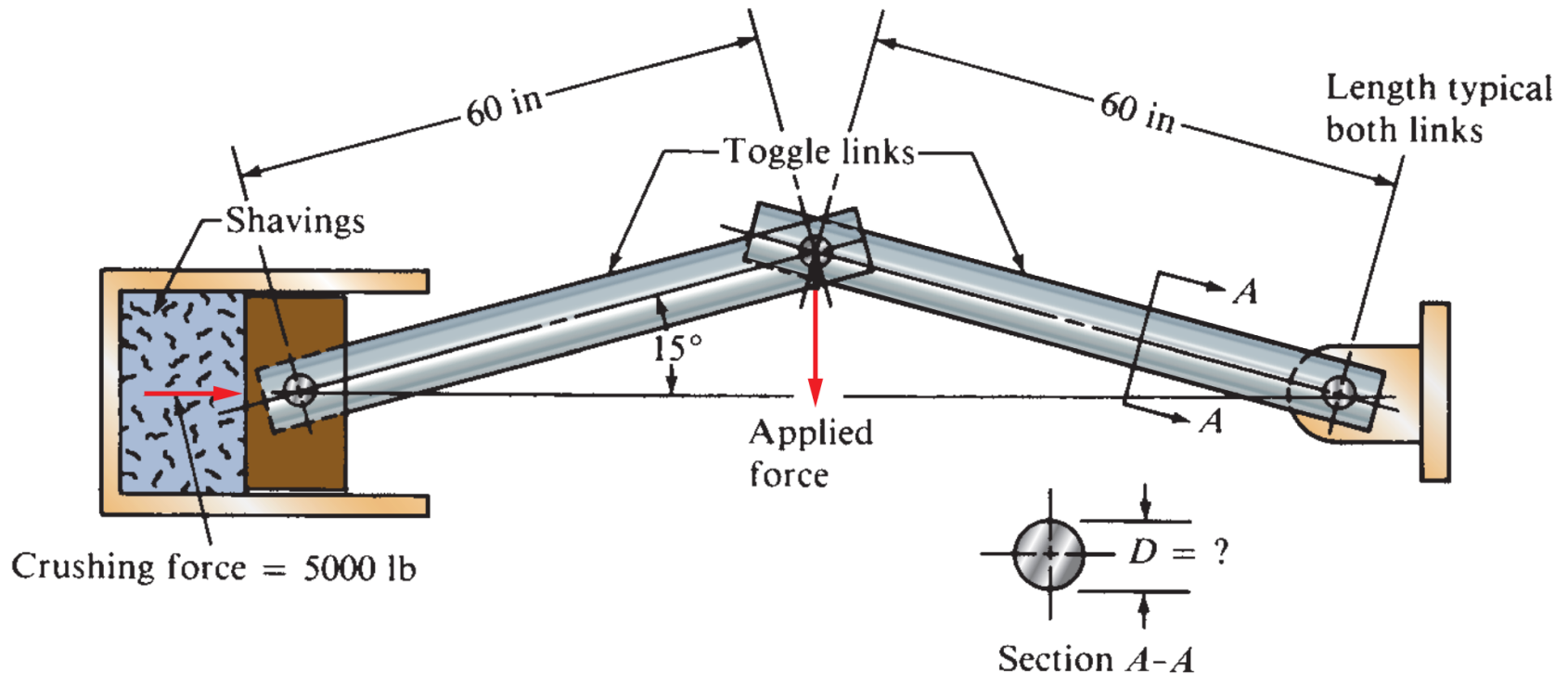
From **Appendix Figure A18-1**, we estimate the stress concentration factor $K_t = 2.0$

Finally, using Goodman criterion,

$$\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t \sigma_a}{s_n'} = \frac{39298}{221000} + \frac{2.0 \times 12030}{46800} = 0.6919$$

$N = 1.45$

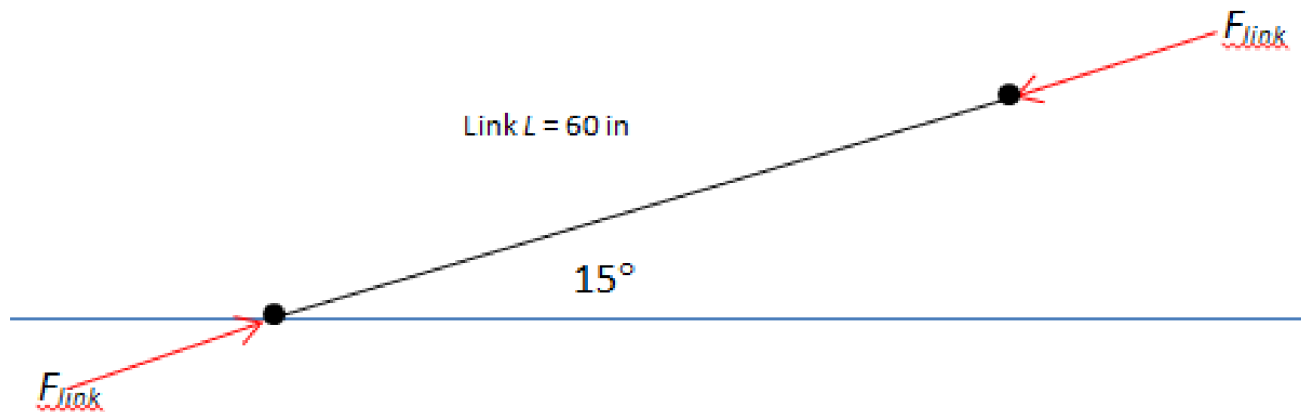
6. A toggle device is being used to compact scrap steel shavings. Design a suitable diameter for the **left** link of the toggle to be steel, SAE 5160 OQT 1000, with a circular cross section and pinned ends. The force P required to crush the shavings is 5000 lb. Use safety factor $N = 3.50$. (10' in total)



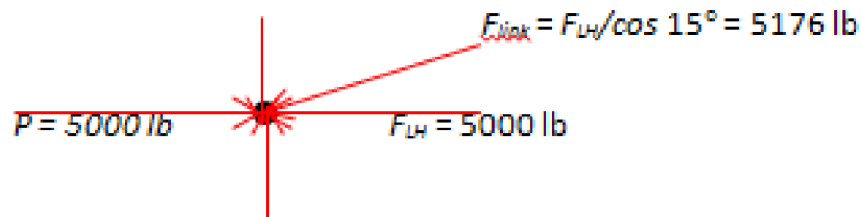
Solutions to Problem #6:

Use Free-Body-Diagram (see below) to get the axial force in the links:

$$F_{\text{link}} = \frac{P}{\cos 15^\circ} = \frac{5000}{\cos 15^\circ} = 5176 \text{ lb}$$



Free-body diagram for lower pin:



Solutions to Problem #6, cont'd:

The material is SAE 5160 OQT 1000. From Appendix 3, we know that the yield strength is $s_y = 151$ ksi and modulus of elasticity $E = 30$ Mpsi.

Assume column is long:

From Equation (6-11) in the textbook, the required diameter for a long, solid circular column is:

$$D = \left[\frac{64NP_a(KL)^2}{\pi^3 E} \right]^{1/4} = \left[\frac{64(3.5)(5176)(1.0 \times 60)^2}{\pi^3 (30 \times 10^6)} \right]^{1/4} = 1.455 \text{ in}$$

The radius of gyration can now be determined (according to Appendix 1):

$$r = \frac{D}{4} = \frac{1.455}{4} = 0.364 \text{ in}$$

$$\text{The corresponding Slenderness Ratio } S_r = \frac{KL}{r} = \frac{1.0 \times 60}{0.364} = 164.8$$

Solutions to Problem #6, cont'd:

The transition slenderness ratio $C_c = S_{rT} = \sqrt{\frac{2\pi^2 E}{s_y}} = \sqrt{\frac{2\pi^2 (30 \times 10^6)}{151 \times 10^3}} = 62.6$.

Since $S_r > C_c$, our assumption of long column is correct.

Therefore, the suitable diameter of the left link is

$$D = 1.455 \text{ in}$$